SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – SHORT ANSWER TEST (30 Questions)

Course Code:	ITA0502	Course Title:	COMPUTER VISION
CO Assessed:	CO1 – Imitation (BL3)	Assessment:	Assessment Tool 1 – Short Answer Test
Weightage:	40% of CIA	Total Marks:	50 (10 Questions × 5 Marks)
CO4 Statement:	*Apply the fundamental concepts, image formation models, and processing techniques for analyzing Computer Vision systems.* (BTL 3)		
Student Name:	Thulasi . I	Reg. No.:	192421032
Section / Batch:	2024 - Slot C	Date:	

Code of Conduct

I Thulasi . I (192421032) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: ____________________

Instructions

- This test contains 6 questions, each carrying 5 marks. Total: 30 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus	Q Nos	Marks
Overview of Computer Vision	Definition, objectives, workflow, and importance of Computer Vision	1	5
Features of Computer Vision	Feature extraction, object recognition, and image understanding	2	5
Levels of Computer Vision and Image Processing	Low-level, mid-level, and high-level image processing operations	3	5
Digital Image Fundamentals	Pixels, image representation, grayscale/color images, and image resolution	4,10	10
Image Sensing and Acquisition	Image sensors, acquisition process, and imaging devices	5	5
Sampling and Quantization	Sampling, quantization, and digital image formation	6	5
Image Formation Models	Perspective projection, illumination, and image formation process	7	5
Applications of Image Processing and Computer Vision	Healthcare, autonomous vehicles, surveillance, agriculture, robotics, and industrial automation	8,9	10
		GRAND TOTAL:	50 Marks

1. How does Computer Vision enable computers to interpret and understand visual information?

a) What are the main objectives of Computer Vision?
b) Mention any two real-world applications of Computer Vision.

2. What are the key features of Computer Vision that make it suitable for intelligent image analysis?

a) Define feature extraction.
b) How does object recognition improve Computer Vision systems?

3. How are the different levels of image processing used in Computer Vision?

a) What is low-level image processing?
b) Differentiate between mid-level and high-level image processing.

4. How is a digital image represented and what are its fundamental components?

a) Define a pixel.
b) How does image resolution affect image quality?

5. How does image sensing and acquisition convert a real-world scene into a digital image?

a) What is the role of an image sensor?
b) Name any two image acquisition devices.

6. How do sampling and quantization contribute to the formation of a digital image?

a) Define sampling.
b) What is the purpose of quantization in image processing?

7. How does the image formation model describe the process of capturing images?

a) What is perspective projection?
b) What is the role of illumination in image formation?

8. How are image processing and Computer Vision related, and how do they differ?

a) Define image processing.
b) State two differences between image processing and Computer Vision.

9. How is Computer Vision applied in solving real-world problems across different domains?

a) Explain one application in healthcare.
b) Explain one application in autonomous vehicles.

10. How do digital image fundamentals support Computer Vision applications?

a) Why are pixels considered the basic building blocks of digital images?
b) How do sampling and image resolution influence image analysis?

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks	
Understanding of Concepts	50%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions		
Presentation	50%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation, lacks depth	Poor or unclear explanation		

Faculty In-charge	Course Coordinator	Head of Department
Signature: ______________	Signature: ______________	Signature: ______________
Date: ______________	Date: ______________	Date: ______________

Short Answer Test

I. Thulasi
192421032.

① Computer vision uses Algorithms and AI technique to analyze images and videos and exract meaning information.

4/5

a) Objectives :

* image recognition
* object detection and recognition
* Scene analysis.

b) real world :

* Face recognition
* self driving cars.

② * features extraction
* pattern recognition
* object detection

a) feature extraction :

feature extraction is the process of identifying important characteristics such as edges. Corners and textures

b) It helps the system to identify and classify, and recognition the videos and images.

③ different levels of image processing:

* high level
* mid level
* low level.

a) low-level image processing:

low-level image processing is used to remove noise, and perform basic operations as image filtering and enhancement

b) Diff/B mid and high level

* mid level feature used to extraction and segmentation.
* high level is used to decision making and object recognition.

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④ A digital image is represented as image as matrix of pixels.

a) pixel: A pixel is the smallest element of a digital image that stores color or intensity information

b) Higher resolution provides more details about more quality and clearness of an image.

sensors capture light from a scene and convert it into digital signals.

(a) Role of an image sensor it converts light energy into electrical signals.

b) * Digital Camera
 * Scanner

⑥ Sampling determines spatial resolution, while quantization determines intensity levels.

(a) Sampling:

Sampling is the process of converting a continuous image into discrete pixels.

(b) purpose of quantization in image processing, it converts continuous intensity value into ~~discrete~~ digital values.

⑦ It explains how light from object is projected onto a camera sensor to form an image.

(a) perspective projection:

It is the process of projecting a 3D scene onto a

2D image plane.

b) Role of illumination in image formation:

Illumination provides a light required to make object visible in the computer

(8) Image processing enhances images, while computer vision interprets and understands them

a) Image processing:

Image processing is the manipulation and enhancement of digital images

b) D/B image processing and computer vision

* Image processing focuses on image enhancement; computer vision focuses on understanding images.

* Image processing produces improved images. Computer vision produces decisions or interpretations.

(9) a) application in healthcare:- medical image analysis for disease detection.

(9) one application in autonomous vehicles : Lane detection and obstacle recognition.

(10) They provide the basic representations and information needed for image analysis.

(a) Because every digital image is composed of pixels containing image information.

(b) Higher sampling and resolution provide and improve image details and analysis accuracy.